Circles and Parabolas

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Circles**

Write the standard-form equation of the circle with center $(3, -2)$ and radius 5.

$(x-3)^2 + (y+2)^2 = 25$.

2 **[Foundational]** • **Target: Circles**

Consider the circle $x^2 + y^2 - 6x + 4y - 12 = 0$.

a Write the equation in standard form by completing the square. $(x-3)^2 + (y+2)^2 = 25$.

b State the center and radius. Center $(3, -2)$, radius 5.

(a) $(x-3)^2 + (y+2)^2 = 25$.; (b) Center $(3, -2)$, radius 5.

3 **[Foundational]** • **Target: Circles**

Determine whether the point $(6, 2)$ lies inside, on, or outside the circle $(x-3)^2 + (y+2)^2 = 25$.

$(6-3)^2 + (2+2)^2 = 9 + 16 = 25$, so the point lies on the circle.

4 **[Foundational]** • **Target: Parabolas with vertex at the origin**

For the parabola $y^2 = 8x$, find:

a the focus $(2, 0)$

b the directrix $x = -2$

c the direction of opening opens right

(a) $(2, 0)$; (b) $x = -2$; (c) opens right

5 **[Foundational]** • **Target: Parabolas with vertex at the origin**

A parabola has vertex at the origin, opens upward, and passes through the point $(4, 2)$.

a Find its equation in the form $x^2 = 4py$. $16 = 4p(2)$ gives $p = 2$, so $x^2 = 8y$.

b State the coordinates of the focus. $(0, 2)$.

(a) $16 = 4p(2)$ gives $p = 2$, so $x^2 = 8y$.; (b) $(0, 2)$.

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6 [Foundational] • Target: Parabolas with vertex at the origin

A satellite dish has a parabolic cross-section 2 m wide and 0.5 m deep. Taking the vertex at the origin opening upward, find the distance from the vertex to the focus (where the receiver is placed).

The edge point $(1, 0.5)$ lies on $x^2 = 4py$: $1 = 4p(0.5) = 2p$, so $p = 0.5$ m. The receiver sits 0.5 m above the vertex.

7 [Foundational] • Target: Parabolas with vertex not at the origin

In a quantitative modeling scenario, for the parabola $(y-2)^2 = 12(x+1)$, find:

a the vertex $(-1, 2)$

b the value of p $4p = 12$, so $p = 3$

c the focus $(2, 2)$

d the directrix $x = -4$

(a) $(-1, 2)$; (b) $4p = 12$, so $p = 3$; (c) $(2, 2)$; (d) $x = -4$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Parabolas with vertex not at the origin

Write the equation of the parabola with vertex $(2, -3)$, opening downward, with $|p| = 1$.

Opening downward: $(x-2)^2 = -4(y+3)$.

9 [Intermediate] • Target: Parabolas with vertex not at the origin

Complete the square to write $y = x^2 - 6x + 5$ in vertex form, and state the vertex.

$y = (x-3)^2 - 4$. Vertex: $(3, -4)$.

10 [Intermediate] • Target: Mixed practice: circles and parabolas together

A circle and a parabola are given by $x^2 + y^2 = 25$ and $y = x^2 - 5$. Find their points of intersection.

Substitute $x^2 = y + 5$ into the circle: $(y+5) + y^2 = 25$, so $y^2 + y - 20 = 0$, giving $(y+5)(y-4) = 0$: $y = -5$ or $y = 4$. For $y = -5$: $x^2 = 0$, point $(0, -5)$. For $y = 4$: $x^2 = 9$, points $(\pm 3, 4)$. Three intersection points: $(0, -5)$, $(3, 4)$, $(-3, 4)$.

11 [Intermediate] • Target: Mixed practice: circles and parabolas together

A circular satellite orbit has the equation $x^2 + y^2 = (6400 + 500)^2$ (Earth radius plus altitude, in km). Find the radius of this orbit.

Radius $= 6400 + 500 = 6900$ km.

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12 [Intermediate] • Target: Graphing circles and parabolas

The parabola $y = x^2 - 2x - 3$ is shown. Use the graph to estimate its vertex and x-intercepts, then verify algebraically.

From the graph, x-intercepts near $x = -1, 3$ and vertex near $(1, -4)$. Algebraically: $y = (x-1)^2 - 4$, vertex $(1, -4)$; setting $y = 0$: $(x-1)^2 = 4 \Rightarrow x = -1, 3$ ✓.

13 [Intermediate] • Target: Tangent lines to circles

Find the equation of the tangent line to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$, using the fact that the tangent is perpendicular to the radius at that point.

Radius slope from origin to $(3, 4)$: $\frac{4}{3}$. Tangent slope (perpendicular): $-\frac{3}{4}$. Tangent line: $y - 4 = -\frac{3}{4}(x - 3)$, i.e. $y = -\frac{3}{4}x + \frac{25}{4}$.

14 [Intermediate] • Target: The reflective property of parabolas

Explain the reflective property of a parabola that makes it useful for satellite dishes and headlights, in terms of the focus.

Any ray traveling parallel to a parabola's axis of symmetry reflects off the interior surface and passes through the focus (or, conversely, rays originating at the focus reflect out parallel to the axis) — this is why satellite dishes concentrate incoming signals at the focus, and headlights placed at the focus project a parallel beam.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Finding a circle through three points • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Explain the general strategy for finding the equation of a circle passing through three given (non-collinear) points, without fully solving a specific example.

Substitute each of the three points into the general circle equation $x^2 + y^2 + Dx + Ey + F = 0$, producing a system of three linear equations in D, E, F. Solving this system gives the circle's equation, which can then be converted to standard form by completing the square.

16 [Challenging] • Target: Distance from a point to the focus (parabola definition) • Cross-Lesson: 9.02 Geometric Sequences and Series

State the focus-directrix definition of a parabola, and use it to verify that the point $(4, 4)$ lies on $y^2 = 4x$ (focus $(1, 0)$, directrix $x = -1$).

A parabola is the set of points equidistant from the focus and directrix. At $(4, 4)$: distance to focus $(1, 0)$ is $\sqrt{(4-1)^2 + 4^2} = \sqrt{9 + 16} = 5$. Distance to directrix $x = -1$ is $4 - (-1) = 5$. Equal, confirming the point lies on the parabola.

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17 [Challenging] • Target: Systems involving circles • Cross-Lesson: 9.03 The Binomial Theorem

Find the points of intersection of the circles $x^2 + y^2 = 25$ and $(x - 6)^2 + y^2 = 25$.

Subtracting the equations: $x^2 - (x - 6)^2 = 0 \Rightarrow x^2 - (x^2 - 12x + 36) = 0 \Rightarrow 12x - 36 = 0 \Rightarrow x = 3$. Substituting into the first: $9 + y^2 = 25 \Rightarrow y = \pm 4$. Intersection points: $(3, 4)$ and $(3, -4)$.

18 [Challenging] • Target: Finding a circle through three points • Cross-Lesson: 1.01 Functions and Function Notation

Find the equation of the circle in standard form that passes through the three points $(0, 0)$, $(6, 0)$, and $(0, 8)$.

Step 1: Recognize right triangle: Points $(6, 0)$, $(0, 0)$, and $(0, 8)$ form a right triangle with right angle at $(0, 0)$. By Thales's theorem, the hypotenuse connecting $(6, 0)$ and $(0, 8)$ is the circle's diameter. Step 2: Find center: The center (h, k) is the midpoint of the diameter: $(h, k) = \left(\frac{6+0}{2}, \frac{0+8}{2}\right) = \mathbf{(3, 4)}$. Step 3: Find radius: The radius squared is the distance from center $(3, 4)$ to the origin $(0, 0)$: $r^2 = 3^2 + 4^2 = 9 + 16 = 25$. The standard eq.

19 [Challenging] • Target: Distance from a point to the focus (parabola definition) • Cross-Lesson: 1.02 Domain and Range

Find the equation of the tangent line to the circle $x^2 + y^2 = 25$ at the point $P(3, 4)$.

Step 1: Radius slope: The center is the origin $O(0, 0)$. The slope of the radius connecting $(0, 0)$ and $(3, 4)$ is: $m_{\text{radius}} = \frac{4-0}{3-0} = \frac{4}{3}$. Step 2: Tangent line slope: Since the tangent is perpendicular to the radius: $m_{\text{tangent}} = -\frac{3}{4}$. Step 3: Point-slope equation: Using point $P(3, 4)$:

$y - 4 = -\frac{3}{4}(x - 3) \Rightarrow 4(y - 4) = -3(x - 3) \Rightarrow 4y - 16 = -3x + 9 \Rightarrow \mathbf{3x + 4y = 25}$.

20 [Challenging] • Target: Systems involving circles • Cross-Lesson: 1.03 Average Rate of Change

A tangent from an external point P touches a circle at T. If the distance from P to the circle's centre O is 13 and the radius is 5, find the length of the tangent segment PT.

Step 1: Radius OT is perpendicular to tangent line PT at point of tangency T, forming right triangle ΔOPT with hypotenuse $OP = 13$ and leg $OT = 5$. Step 2: Apply the Pythagorean theorem: $PT = \sqrt{OP^2 - OT^2} = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = \mathbf{12}$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Circles

A circle $\mathcal{C}$ with center $C(0, k)$ on the positive y-axis passes through the focus F of the parabola $x^2 = 4y$. If $\mathcal{C}$ is tangent to the parabola at two distinct points, P and Q, what is the area of quadrilateral $PFQC$?

a. $8\sqrt{3}$ ✓

b. $4\sqrt{3}$

c. $16\sqrt{3}$

d. $6\sqrt{3}$

1. **Identify the focus and circle equation:** The standard form of the parabola is $x^2 = 4py$, so with $4p = 4$, we have $p = 1$. Thus, the focus is $F(0, 1)$. The circle is centered at $C(0, k)$ with $k > 1$ and passes through $F(0, 1)$, so its radius is $R = k - 1$. The equation of $\mathcal{C}$ is: $x^2 + (y - k)^2 = (k - 1)^2$ 2. **Find the tangency condition:** Substitute $x^2 = 4y$ from the parabola into the circle's equation: $4y + y^2 - 2ky + k^2 = k^2 - 2k + 1$ $y^2 - 2(k - 2)y + (2k - 1) = 0$ For.

22 [Foundational] • Target: Parabolas with vertex at the origin

In the diagram below, a circle has its center at the origin $(0, 0)$ and passes through the point $(3, 4)$. What is the radius of the circle?

a. **5** ✓

b. 7

c. 25

d. 12

The equation of a circle centered at the origin is $x^2 + y^2 = r^2$, where r is the radius. Since the circle passes through the point $(3, 4)$, we can substitute these coordinates into the equation: $3^2 + 4^2 = r^2$. This simplifies to $9 + 16 = r^2$, so $r^2 = 25$. Taking the square root of both sides, we get $r = 5$ (since the radius must be positive). Therefore, the radius of the circle is 5.

23 [Foundational] • Target: Parabolas with vertex not at the origin

A parabola has its vertex at the origin $(0, 0)$ and passes through the point $(2, -4)$. If the directrix of the parabola is a horizontal line, what is the equation of the directrix?

a. **y = 4** ✓

b. y = -4

c. x = 4

d. x = -4

The vertex of the parabola is at $(0, 0)$. Since the directrix is a horizontal line, the axis of symmetry is vertical, and the equation of the parabola is of the form $x^2 = 4py$. The parabola passes through the point $(2, -4)$. Substituting these coordinates into the equation, we get $2^2 = 4p(-4)$, which simplifies to $4 = -16p$. Solving for p, we find $p = \frac{4}{-16} = -\frac{1}{4}$. The equation of the parabola is $x^2 = 4(-\frac{1}{4})y$, or $x^2 = -y$. The directrix of a parabola of the form $x^2 = 4py$ i.

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24 [Foundational] • Target: Mixed practice: circles and parabolas together

The graph of the quadratic function $f(x) = x^2 - bx + c$ is shown below. The vertex of the parabola is at $(2, -3)$, and it passes through the point $(0, 1)$. Determine the value of the parameter b.

a. **4** ✓

b. 3

c. -4

d. -3

The standard form of a quadratic function is $f(x) = ax^2 + bx + c$. In this problem, $a = 1$. The vertex form of a quadratic function is $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. Given the vertex is $(2, -3)$ and $a = 1$, the function can be written as $f(x) = 1(x - 2)^2 - 3$. Expanding this, we get $f(x) = (x^2 - 4x + 4) - 3 = x^2 - 4x + 1$. We are given that $f(x) = x^2 - bx + c$. By comparing the coefficients, we see that $-b = -4$, which implies $b = 4$, and $c = 1$. We can verify this by checking if the.

25 [Foundational] • Target: Graphing circles and parabolas

The quadratic function $f(x) = ax^2 + bx + c$ is graphed in the standard (x, y) coordinate plane below. The vertex of the parabola is at $(3, -6)$, and the graph passes through the point $(1, 2)$. What is the value of b ?

a. -12 ✓

b. -6

c. 6

d. 12

1. Write the quadratic function in vertex form: $f(x) = a(x - h)^2 + k$. 2. Given the vertex $(h, k) = (3, -6)$, the function is $f(x) = a(x - 3)^2 - 6$. 3. Use the given point $(1, 2)$ on the graph to find a: $2 = a(1 - 3)^2 - 6$ $2 = 4a - 6 \Rightarrow 4a = 8 \Rightarrow a = 2$. 4. Expand the equation to standard form $ax^2 + bx + c$: $f(x) = 2(x^2 - 6x + 9) - 6 = 2x^2 - 12x + 18 - 6 = 2x^2 - 12x + 12$. 5. Comparing coefficients gives $b = -12$. Common error: A student who ignores the scale factor a might cal.

26 [Foundational] • Target: Tangent lines to circles

The graph of a quadratic function $f(x) = ax^2 + bx + c$ is shown below. The vertex of the parabola is at $(2, -4)$, and the function passes through the point $(4, 0)$. Determine the value of the coefficient a.

a. **1** ✓

b. -1

c. 2

d. -2

The vertex form of a quadratic function is $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. Given the vertex is $(2, -4)$, the function can be written as $f(x) = a(x - 2)^2 - 4$. We are also given that the graph passes through the point $(4, 0)$. Substituting these coordinates into the equation: $0 = a(4 - 2)^2 - 4$ $0 = a(2)^2 - 4$ $0 = 4a - 4$ $4a = 4$ $a = 1$. Therefore, the value of the coefficient a is 1.

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27 [Foundational] • Target: The reflective property of parabolas

Consider the function $g(x) = ax^2 + bx + c$. The graph of $y = g(x)$ is a parabola that passes through the points $(1, 5)$, $(2, 12)$, and $(3, 21)$. If the vertex of the parabola is at (h, k), what is the value of h?

a. $\frac{5}{2}$ ✓

b. 2

c. $-\frac{5}{2}$

d. -2

Let the quadratic function be $g(x) = ax^2 + bx + c$. We are given three points that lie on the graph of $y = g(x)$: 1. $(1, 5) \Rightarrow a(1)^2 + b(1) + c = 5 \Rightarrow a + b + c = 5$ (Equation 1) 2. $(2, 12) \Rightarrow a(2)^2 + b(2) + c = 12 \Rightarrow 4a + 2b + c = 12$ (Equation 2) 3. $(3, 21) \Rightarrow a(3)^2 + b(3) + c = 21 \Rightarrow 9a + 3b + c = 21$ (Equation 3) Subtract Equation 1 from Equation 2: $(4a + 2b + c) - (a + b + c) = 12 - 5 \Rightarrow 3a + b = 7$ (Equation 4) Subtra.

28 [Foundational] • Target: Finding a circle through three points

In the standard (x, y) coordinate plane, the line $y = 2x - 3$ intersects the parabola $y = x^2 - 6x + c$ at two distinct points, P and Q. If the distance between P and Q is $4\sqrt{5}$, what is the value of the constant c?

a. -7

b. 9 ✓

c. 12

d. 15

To find the points of intersection, set the two equations equal: $x^2 - 6x + c = 2x - 3$ Rearranging into standard quadratic form gives: $x^2 - 8x + (c + 3) = 0$ Let the roots of this equation be x_1 and x_2. By Vieta's formulas, the sum and product of the roots are: $x_1 + x_2 = 8$ and $x_1x_2 = c + 3$ The horizontal distance between the intersection points is given by $|x_2 - x_1|$. We can calculate $(x_2 - x_1)^2$ using the identity.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Distance from a point to the focus (parabola definition)

Consider the function $f(x) = ax^2 + bx + c$. The graph of $y = f(x)$ is a parabola that passes through the points $(-1, 5)$ and $(2, -7)$. If the vertex of the parabola is located at $x = -1/2$, find the value of c.

a. **1 ✓**

b. 2

c. 3

d. 4

The vertex of a parabola given by $f(x) = ax^2 + bx + c$ occurs at $x = -b/(2a)$. We are given that the vertex occurs at $x = -1/2$, so $-b/(2a) = -1/2$, which implies $b = a$. The parabola passes through $(-1, 5)$, so $f(-1) = a(-1)^2 + b(-1) + c = a - b + c = 5$. Since $b = a$, this becomes $a - a + c = 5$, which simplifies to $c = 5$. The parabola also passes through $(2, -7)$, so $f(2) = a(2)^2 + b(2) + c = 4a + 2b + c = -7$. Substituting $b = a$ and $c = 5$ into this equation gives $4a + 2a + 5 = -7$, so.

30 [Intermediate] • Target: Systems involving circles

A circle has center $(6, 6)$ and radius 2. Find the shortest distance from the point $(16, 30)$ to the circle.

a. 6

b. 38

c. **24 ✓**

d. 29

Distance from point to center $= \sqrt{10^2 + 24^2} = 26$. Since $26 > 2$, the point is outside; shortest distance to the circle $= 26 - 2 = 24$.

31 [Intermediate] • Target: Circles

A circle has center $(4, -2)$ and passes through $(13, 10)$. Find the slope of the tangent line to the circle at $(13, 10)$.

a. 5/4

b. 1/4

c. **-3/4 ✓**

d. -11/4

Radius slope $= \frac{10+2}{13-4} = 4/3$. Tangent $\perp$ radius, so tangent slope $= -\frac{1}{4/3} = -3/4$.

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32 [Intermediate] • Target: Parabolas with vertex at the origin

Find all points on the parabola $y^2 = 4x$ equidistant from the focus and from the point $(6, 0)$.

a. $(4, 4)$ only

b. $(1, 2)$ only

c. $(0, 0)$

d. $(1, 2), (1, -2)$ **and** $(9, 6), (9, -6)$ ✓

Distance from focus $(1, 0)$ equals distance from $(6, 0)$: condition yields two pairs.

33 [Intermediate] • Target: Parabolas with vertex not at the origin

Two tangent lines from an external point P to a conic: the segment from P to each tangent point has the property that the two tangent lengths are equal for a:

a. Hyperbola

b. Ellipse

c. **Circle** ✓

d. Parabola

Equal tangent lengths from external point only for a circle.

34 [Intermediate] • Target: Mixed practice: circles and parabolas together

The graph of a third-degree polynomial function $P(x)$ is shown in the standard (x, y) coordinate plane below. The graph crosses the x-axis at $(-1, 0)$, is tangent to the x-axis at $(2, 0)$, and passes through the point $(0, -2)$. What is the value of $P(-2)$?

a. -16

b. -8

c. 4

d. **8** ✓

To find $P(-2)$, we first determine the algebraic formula for the polynomial $P(x)$. 1. **Identify the factors and their multiplicities:** - The graph crosses the x-axis at $x = -1$, meaning $(x + 1)$ is a factor with an odd multiplicity. - The graph is tangent to the x-axis at $x = 2$, meaning $(x - 2)$ is a factor with an even multiplicity (at least 2). - Since $P(x)$ is a third-degree (cubic) polynomial, the sum of the multiplicities must equal 3. Therefore, $(x + 1)$ has multiplicity 1 and $(x - 2)$ has mul.

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35 [Intermediate] • Target: Graphing circles and parabolas

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at two points. What is the y-coordinate of the vertex of this parabola?

a. **-1** ✓

b. 1

c. -2

d. 0

The vertex of a parabola in the form $f(x) = ax^2 + bx + c$ occurs at $x = -\frac{b}{2a}$. For the function $f(x) = x^2 - 4x + 3$, we have $a = 1$ and $b = -4$. Thus, the x-coordinate of the vertex is $x = -\frac{-4}{2(1)} = \frac{4}{2} = 2$. To find the y-coordinate of the vertex, we substitute this value of x back into the function: $f(2) = (2)^2 - 4(2) + 3 = 4 - 8 + 3 = -1$. Therefore, the y-coordinate of the vertex is -1.

36 [Intermediate] • Target: Tangent lines to circles

In the diagram below, points A, B, C, and D lie on the circle. If AB = 6, BP = 8, CD = 4, and AP = x, where P is the intersection of chords AC and BD, find the value of x.

a. **4** ✓

b. 5

c. 6

d. 8

The problem involves the intersecting chords theorem (also known as the power of a point theorem). When two chords intersect inside a circle, the product of the lengths of the segments of one chord is equal to the product of the lengths of the segments of the other chord. Let the chords be AC and BD, intersecting at point P. The theorem states: PA · PC = PB · PD From the problem statement, we are given: PA = 8 PB = 8 PC = 4 We need to find the length of PD, which is denoted by x. Substitute the given va.

37 [Intermediate] • Target: The reflective property of parabolas

The quadratic function $f(x) = ax^2 + bx + c$ has its vertex at $(-1, 5)$. If the graph of $f(x)$ also passes through the point $(2, -13)$, find the value of b.

a. 3

b. -3

c. 6

d. **-6** ✓

The vertex form of a quadratic function is $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. Given the vertex is $(-1, 5)$, we have $f(x) = a(x - (-1))^2 + 5 = a(x + 1)^2 + 5$. The graph passes through the point $(2, -13)$, so we can substitute these values into the equation: $-13 = a(2 + 1)^2 + 5$. This simplifies to $-13 = a(3)^2 + 5$, so $-13 = 9a + 5$. Subtracting 5 from both sides gives $-18 = 9a$, which means $a = -2$. Now, we expand the vertex form to find the standard form $f(x) = ax^2 + bx + c$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Finding a circle through three points • Cross-Lesson: 1.01 Functions and Function Notation

The diagram shows the graph of the function $y = f(x)$. The graph is transformed by a sequence of operations. The resulting graph, $y = g(x)$, is also shown. The function $f(x)$ is given by $f(x) = ax^2 + bx + c$, where $a, b,$ and c are constants. The vertex of the graph of $y = f(x)$ is at $(2, -1)$, and it passes through the point $(4, 3)$. The graph of $y = g(x)$ is obtained by translating the graph of $y = f(x)$ by $(-1\ 3)$ followed by a reflection in the y-axis. Find the values of $a, b,$ and c.

a. $a = 1, b = -4, c = 3$ ✓

b. $a = 1, b = 4, c = -5$

c. $a = 2, b = -8, c = 7$

d. $a = -1, b = 4, c = -5$

The vertex of the parabola $y = ax^2 + bx + c$ is at $\left(-\frac{b}{2a}, f\left(-\frac{b}{2a}\right)\right)$. We are given that the vertex is at $(2, -1)$. Therefore, $-\frac{b}{2a} = 2$, which implies $b = -4a$. The vertex form of the quadratic equation is $y = a(x - h)^2 + k$, where (h, k) is the vertex. So, $y = a(x - 2)^2 - 1$. The graph passes through $(4, 3)$. Substituting this point into the equation: $3 = a(4 - 2)^2 - 1 \Rightarrow 3 = a(2)^2 - 1 \Rightarrow 3 = 4a - 1 \Rightarrow 4a = 4 \Rightarrow a = 1$. Si.

39 [Challenging] • Target: Distance from a point to the focus (parabola definition) • Cross-Lesson: 1.02 Domain and Range

The graph of $y = f(x)$ is a curve passing through the points A(1, 2), B(4, 2), and C(2, 5). This graph is then subjected to a sequence of transformations. First, it is reflected in the line $y = x$. Let the image of the points A, B, and C under this reflection be A', B', and C' respectively. Secondly, the graph of the reflected curve is translated by the vector $(-3; 0)$. Let the final image of the points A', B', and C' after this translation be A'', B'', and C''. Given that the original curve is a parabola with its axis of symmetry parallel to the y-axis, and passes through A, B, and C, find the coordinates of C''.

a. (-3, 2)

b. (-2, -3)

c. **(2, -3)** ✓

d. (-3, -2)

The original points are A(1, 2), B(4, 2), and C(2, 5). The curve $y = f(x)$ is a parabola. Since A and B have the same y-coordinate, the axis of symmetry is $x = \frac{1+4}{2} = 2.5$. The equation of the parabola can be written as $y = a(x - 2.5)^2 + k$. Substituting C(2, 5): $5 = a(2 - 2.5)^2 + k \Rightarrow 5 = 0.25a + k$. Step 1: Reflection in the line $y = x$. This transformation swaps the x and y coordinates. A'(2, 1), B'(2, 4), C'(5, 2). Step 2: Translation by the vector $(-3; 0)$. This subtracts 3.

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40 [Challenging] • Target: Systems involving circles • Cross-Lesson: 1.03 Average Rate of Change

In the standard (x, y) coordinate plane, a circle has its center at $(-3, 5)$ and passes through the point $(1, 8)$. What is the radius of this circle?

a. 5 ✓

b. 7

c. $\sqrt{13}$

d. 25

The radius r of a circle is the distance from its center to any point on the circle. Using the distance formula between the center $(-3, 5)$ and the point on the circle $(1, 8)$: $r = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ $r = \sqrt{(1-(-3))^2 + (8-5)^2} = \sqrt{4^2 + 3^2} = \sqrt{16+9} = \sqrt{25} = 5$ Thus, the radius of the circle is 5. Common misconceptions: - Choice B (7) mistakenly adds the coordinate differences directly: $4 + 3 = 7$. - Choice C $(\sqrt{13})$ makes a sign error in subtracting the.

41 [Challenging] • Target: Circles • Cross-Lesson: 2.01 Right-Triangle Trigonometry

In the standard (x, y) coordinate plane, the lines with equations $2x - y = 7$ and $x + y = 5$ intersect at the point (a, b). What is the value of a?

a. 1

b. 3

c. **4** ✓

d. 5

The point of intersection (a, b) satisfies both linear equations simultaneously: 1) $2a - b = 7$ 2) $a + b = 5$ To solve for a, add the two equations together to eliminate b: $(2a - b) + (a + b) = 7 + 5$ $3a = 12$ $a = 4$ Substituting $a = 4$ into the second equation gives $4 + b = 5$, which yields $b = 1$. Thus, the lines intersect at $(4, 1)$, and the value of a is 4. - A common misconception is solving for $b = 1$ and selecting 1 (Choice A). - Another error is an arithmetic mistake in dividing 12 by 3.

42 [Challenging] • Target: Parabolas with vertex at the origin • Cross-Lesson: 2.02 Radian Measure and Arc Length

In the circle shown below, chords AB and CD intersect at point P. If $PA = 4$, $PB = 6$, and $PC = 3$, what is the length of segment $\overline{PD}$?

a. 5

b. 7

c. **8** ✓

d. 9

By the Intersecting Chords Theorem (Power of a Point Theorem), when two chords AB and CD intersect at an interior point P, the products of their segments are equal: $PA \cdot PB = PC \cdot PD$ Substitute the given values into the equation: $(4)(6) = (3)(PD)$ $24 = 3 \cdot PD$ $PD = \frac{24}{3} = 8$ Common error analysis: - Adding the segment lengths instead of multiplying ($PA + PB = PC + PD \Rightarrow 4 + 6 = 3 + PD \Rightarrow PD = 7$) yields 7, which fails because the segments satisfy a multiplicative re.

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43 [Challenging] • Target: Parabolas with vertex not at the origin • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

The graph of the function $f(x) = a\sqrt{x-h} + k$ is shown below. The function passes through the points $(5, 10)$ and $(13, 16)$. Determine the value of a and the coordinates of the vertex (h, k).

a. $a = 3, (h, k) = (1, 1)$

b. $a = 2, (h, k) = (1, 4)$

c. $a = 3, (h, k) = (1, 4)$ ✓

d. $a = 2, (h, k) = (1, 1)$

The vertex of the graph is (h, k). From the graph, we can infer that the vertex is at $(1, 4)$. Thus, $h = 1$ and $k = 4$. The function can be written as $f(x) = a\sqrt{x-1} + 4$. We are given that the graph passes through the point $(5, 10)$. Substituting these values into the function: $10 = a\sqrt{5-1} + 4$. This simplifies to $10 = a\sqrt{4} + 4$, so $10 = 2a + 4$. Subtracting 4 from both sides gives $6 = 2a$, which means $a = 3$. Therefore, $a = 3$ and the vertex is $(h, k) = (1, 4)$.

44 [Challenging] • Target: Mixed practice: circles and parabolas together • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

In the diagram, two chords AB and CD intersect at point P inside a circle. If $PA = 6$, $PB = 8$, and $PC = 4$, what is the length of PD?

a. 6

b. 8

c. **12** ✓

d. 10

By the Power of a Point theorem (Intersecting Chords Theorem), for two chords intersecting inside a circle, the product of the segments of one chord is equal to the product of the segments of the other chord. Therefore, we have $PA \cdot PB = PC \cdot PD$. Substituting the given values: $6 \cdot 8 = 4 \cdot PD$. This simplifies to $48 = 4 \cdot PD$. Dividing both sides by 4, we get $PD = \frac{48}{4} = 12$.

45 [Challenging] • Target: Graphing circles and parabolas • Cross-Lesson: 3.01 The Unit Circle and Exact Values

Consider a circle with center O. Two chords AB and CD intersect at point P inside the circle. If AP = 3, PB = 8, and CP = 4, determine the length of PD. The diagram shows a circle with intersecting chords.

a. **6** ✓

b. 8

c. 10

d. 12

According to the Power of a Point theorem for intersecting chords, the product of the segments of one chord is equal to the product of the segments of the other chord. Therefore, we have AP · PB = CP · PD. Substituting the given values: 3 · 8 = 4 · PD 24 = 4 · PD PD = $\frac{24}{4}$ PD = 6 Thus, the length of PD is 6.

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